VIDYAVARDHAKA COLLEGE OF ENGINEERING

DEPARTMENT OF CSE
(ARTIFICIAL INTELIGENCE & MACHINE LEARNING)

LAB MANUAL

DEVOPS LABORATORY
20CI77
(ACADEMIC YEAR 2023-24)

INSTITUTIONAL MISSION AND VISION

Vision

Vidyavardhaka College of Engineering shall be a leading institution in engineering and management education enabling individuals for significant contribution to the society.

Mission

1. Provide the best teaching-learning environment through competent staff and excellent infrastructure.
2. Inculcate professional ethics, leadership qualities, communication and entrepreneurial skills to meet the societal needs.
3. Promote innovation through research and development.
4. Strengthen industry-institute interaction for knowledge sharing.

DEPARTMENT OF CSE (AI & ML)

Vision

To develop competent and socially responsible engineers in the domain of Artificial Intelligence & Machine Learning for noteworthy contributions to society.

Mission

1. Provide profound learning through proficient staff and state of the art infrastructure.
2. Promote professional environment to imbibe ethical values and entrepreneurial skills.
3. Encourage research and development by emphasizing innovation.
4. Facilitate Industry-Institute collaboration for mutual benefit.

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PROGRAM SPECIFIC OUTCOMES (PSO)

Students graduating in CSE(Artificial Intelligence & Machine Learning) will be able to:

PSO1: Analyze, Design, Implement and Test Complex Systems by applying suitable algorithms, data structures and object-oriented technology**.**

PSO2: Apply appropriate tools and techniques of Communication and Security to provide reliable solutions.

PSO3: Demonstrate the knowledge of Artificial Intelligence, Machine Learning and Data Engineering by designing and developing intelligent systems

PROGRAM OUTCOMES (PO'S) BASED ON NBA GRADUATE ATTRIBUTES

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering, fundamentals, and an engineering specialization to the solution of complex engineering problems. **PO2: Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional Engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in Multi-disciplinary environments.

PO12: Lifelong learning: Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

List of Experiments

Sl.no.	List Experiments	Page No.
	A-Demonstration	
1	Demonstrate and Create project in local and remote repository using GitBash and GitHub and apply init, status, log, add, commit, push, config, clone and reset commands on repository.	
2	Demonstrate to create a project in remote repository and apply fork, merge, diff, merge conflict, branch and pull request concepts on repository using GitHub.	
3	Demonstrate the process of integration github repository with Jenkins to automate the project execution in CI/CD pipeline.	
	B-Exercise	
4	Create a docker image for an application stored in local repository and run the application using docker image.	
5	Create a process to dockerize the image for mysql database, demonstrate the working of client and server model using docker.	
6	Create and configure Jenkins files for workflow and build of an application and push the image	
	C-Structured Inquiry	
7	Create a maven projects with all dependencies required for the application in CI/CD pipeline.	
8	Integrate communication channel with Jenkins for status of project and also enable email notification for a build.	
9	Apply Xpath to locate elements in an application for automation testing in CI/CD pipeline.	
	D-Open Ended Problem	
10	Create a Maven project and integrate all dependencies, integrate the project with github repository , integrate with docker and docker hub and create Continuous Integration / Continuous Deployment pipeline using Jenkins and enable email notification for status of build.	

DevOps

Git

Git track changes via a distributed version control system. This means that Git can track the state of different versions of your projects while you're developing them. It is distributed because you can access your code files from another computer – and so can other developers.

When you're building an open source project, you'll need a way to document or track your code. This helps make your work organized, and lets you keep track of the changes you've made.

Operation of Git

- Manage projects with **Repositories**
- **Clone** a project to work on a local copy
- Control and track changes with **Staging** and **Committing**
- **Branch** and **Merge** to allow for work on different parts and versions of a project
- **Pull** the latest version of the project to a local copy
- **Push** local updates to the main project

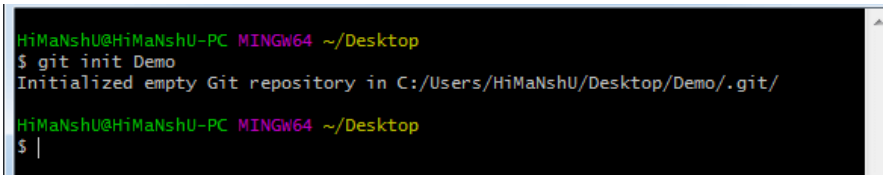
1. Demonstrate and Create project in local and remote repository using GitBash and GitHub and apply init, status, log, add, commit, push, config, clone and reset commands on repository.

Git Init command

This command is used to create a local repository.

Syntax

$ git init Demo



Git status command

The status command is used to display the state of the working directory and the staging area. It also lists the files that you've changed and those you still need to add or commit.

Syntax

$ git status

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/Git-example (master)
$ git status
On branch master
Your branch is based on 'origin/master', but the upstream is gone.
  (use "git branch --unset-upstream" to fixup)

nothing to commit, working tree clean
```

Git log Command

This command is used to check the commit history.

Syntax

$ git log

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/gitexample2 (master)
$ git log
commit 1d2bc037a54eba76e9f25b8e8cf7176273d13af0 (HEAD -> master, origin/master,
origin/HEAD)
Author: ImDwivedi1 <52317024+ImDwivedi1@users.noreply.github.com>
Date:   Fri Aug 30 11:05:06 2019 +0530

    Initial commit
```

By default, if no argument passed, Git log shows the most recent commits first. We can limit the number of log entries displayed by passing a number as an option, such as -3 to show only the last three entries.

Git add command

This command is used to add one or more files to staging (Index) area.

Syntax

To add one file

$ git add Filename

To add more than one file

$ git add*

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/Git-example (master)
$ git add README.md
```

Git commit command

Commit command is used in two scenarios. They are as follows.

Git commit -m

This command changes the head. It records or snapshots the file permanently in the version history with a message.

Syntax

$ git commit -m " Commit Message"

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/Git-example (master)
$ git commit -a -m "Adding the key of c"
[master (root-commit) 758797a] Adding the key of c
 1 file changed, 2 insertions(+)
 create mode 100644 README.md
```

Git push Command

It is used to upload local repository content to a remote repository. Pushing is an act of transfer commits from your local repository to a remote repo. It's the complement to git fetch, but whereas fetching imports commits to local branches on comparatively pushing exports commits to remote branches.

Syntax

$ git push [variable name] master

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/Git-example (mast
er)
$ git push origin master
```

Git config command

This command configures the user. The Git config command is the first and necessary command used on the Git command line.

Syntax

$ git config --global user.name "ImDwivedi1"
$ git config --global user.email "Himanshudubey481@gmail.com"

Git clone command

This command is used to make a copy of a repository from an existing URL. If I want a local copy of my repository from GitHub, this command allows creating a local copy of that repository on your local directory from the repository URL.

Syntax

$ git clone URL

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/Git-example (master)
$ git clone https://github.com/ImDwivedi1/Git-Example.git
Cloning into 'Git-Example'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (3/3), done.
```

Git Reset

The term reset stands for undoing changes. The git reset command is used to reset the changes.

$ git reset --hard

Consider the below output:

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (test2)
$ git reset --hard
HEAD is now at 34c25eb Revert "CSS file "
```

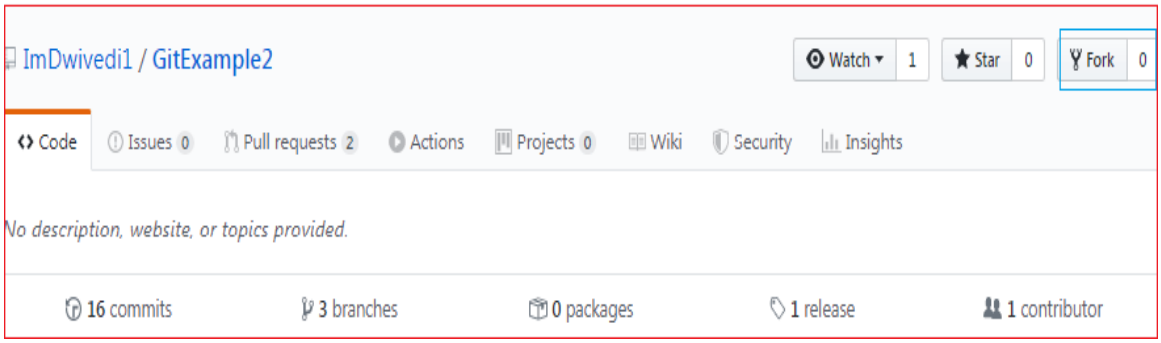
2. Demonstrate to create a project in remote repository and apply fork, merge, diff, merge conflict, branch and pull request concepts on repository using GitHub.

Git Fork

A fork is a rough copy of a repository. Forking a repository allows you to freely test and debug with changes without affecting the original project.

It is a straight-forward process. Steps for forking the repository are as follows:

- Login to the GitHub account.
- Find the GitHub repository which you want to fork.
- Click the Fork button on the upper right side of the repository's page.



Git Merge

In Git, the merging is a procedure to connect the forked history. It joins two or more development history together. The git merge command facilitates you to take the data created by git branch and integrate them into a single branch.

```
$ git merge master
```

See the below

output:

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (test)
$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.

HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$ git merge 2852e020909dfe705707695fd6d715cd723f9540
Updating 4a6693a..2852e02
Fast-forward
 newfile.txt  | 1 +
 newfile1.txt | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 newfile.txt
 create mode 100644 newfile1.txt

HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$
```

Git Diff

It compares the different versions of data sources. The version control system stands for working with a modified version of files. So, the diff command is a useful tool for working with Git.

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (test2)
$ git diff
diff --git a/newfile1.txt b/newfile1.txt
index ade63b7..41a6a9c 100644
--- a/newfile1.txt
+++ b/newfile1.txt
@@ -3,3 +3,4 @@ i am on test2 branch.
 git commit1
 git commit2
 git merge demo
+changes are made to understand the git diff command.
```

Git Merge Conflict

When two branches are trying to merge, and both are edited at the same time and in the same file, Git won't be able to identify which version is to take for changes. Such a situation is called merge conflict.

$ mkdir git-merge-test

$ cd git-merge-test

$ git init .

$ echo "this is some content to mess with" > merge.txt

$ git add merge.txt

$ git commit -am"we are commiting the inital content"

[main (root-commit) d48e74c] we are commiting the inital

content 1 file changed, 1 insertion(+)

create mode 100644 merge.txt

$ git checkout -b new_branch_to_merge_later

$ echo "totally different content to merge later" > merge.txt

$ git commit -am"edited the content of merge.txt to cause a conflict"

[new_branch_to_merge_later 6282319] edited the content of merge.txt to cause a

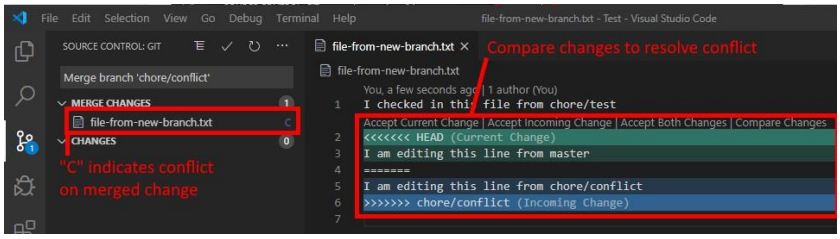
conflict 1 file changed, 1 insertion(+), 1 deletion(-)

$ git merge

new_branch_to_merge_later Auto-

merging merge.txt

CONFLICT (content): Merge conflict in merge.txt



Automatic merge failed; fix conflicts and then commit result

Git Pull

The term pull is used to receive data from GitHub. It fetches and merges changes from the remote server to your working directory. The **git pull command** is used to pull a repository.

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$ git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (3/3), done.
From https://github.com/ImDwivedi1/GitExample2
   f1ddc7c..0a1a475  master     -> origin/master
Updating f1ddc7c..0a1a475
Fast-forward
 design2.css | 6 ++++++
 1 file changed, 6 insertions(+)
 create mode 100644 design2.css

HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$ |
```

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/Demo (master)
$ git pull https://github.com/ImDwivedi1/GitExample2.git
remote: Enumerating objects: 38, done.
remote: Counting objects: 100% (38/38), done.
remote: Compressing objects: 100% (25/25), done.
remote: Total 38 (delta 13), reused 19 (delta 7), pack-reused 0
Unpacking objects: 100% (38/38), done.
From https://github.com/ImDwivedi1/GitExample2
 * branch            HEAD       -> FETCH_HEAD

HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/Demo (master)
$
```

Git Branch

A branch is a version of the repository that diverges from the main working project. It is a feature available in most modern version control systems. A Git project can have more than one branch.

Create Branch

You can create a new branch with the help of the **git branch** command. This command will be used as:

Syntax:

$ git branch **<branch** name**>**

Output:

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$ git branch B1
```

List Branch

You can List all of the available branches in your repository by using the following command.

$ git branch

Output:

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$ git branch
  B1
  branch3
* master

HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$ git branch --list
  B1
  branch3
* master
```

$ git branch -d**<branch** name**>**

Output:

```
HiMaNshU@HiMaNshU-PC MINGW64 ~/Desktop/GitExample2 (master)
$ git branch -d B1
Deleted branch B1 (was 554a122).
```

3. Demonstrate the process of integration github repository with Jenkins to automate the project execution in CI/CD pipeline.

Install Jenkins on Windows

1. Install Java Development Kit (JDK)
2. Download JDK 8 and choose windows 32-bit or 64-bit according to your system configuration.

Click on "accept the license agreement." Set the Path for the Environmental Variable for JDK.

- Go to System Properties. Under the "Advanced" tab, select "Environment Variables."
- Under system variables, select "new." Then copy the path of the JDK folder and paste it in

the corresponding value field. Similarly, do this for JRE.

•Under system variables, set up a bin folder for JDK in PATH variables.

•Go to command prompt and type the following to check if Java has been successfully installed:

2. Download and Install Jenkins

3. Download Jenkins. Under LTS, click on windows.

•After the file is downloaded, unzip it. Click on the folder and install it. Select "finish" once done.

4. Run Jenkins on Localhost 8080

•Once Jenkins is installed, explore it. Open the web browser and type "localhost:8080".

•Enter the credentials and log in. If you install Jenkins for the first time, the dashboard will ask you to install the recommended plugins. Install all the recommended plugins.

5. Jenkins Server Interface

•New Item allows you to create a new project.

•Build History shows the status of your builds.

•Manage System deals with the various configurations of the system.

6. Build and Run a Job on Jenkins

•Select a new item (Name - Jenkins_demo). Choose a freestyle project and click Ok.

•Under the General tab, give a description like "This is my first Jenkins job." Under the "Build Triggers" tab, select add built step and then click on the "Execute Windows" batch command.

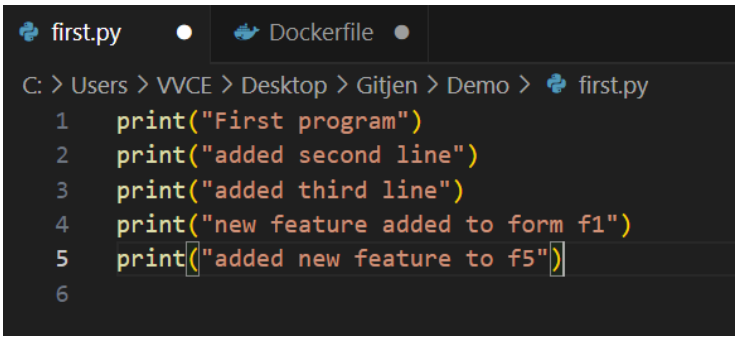
•In the command box, type the following: echo "Hello... This is my first Jenkins Demo: %date%:

%time% ". Click on apply and then save.

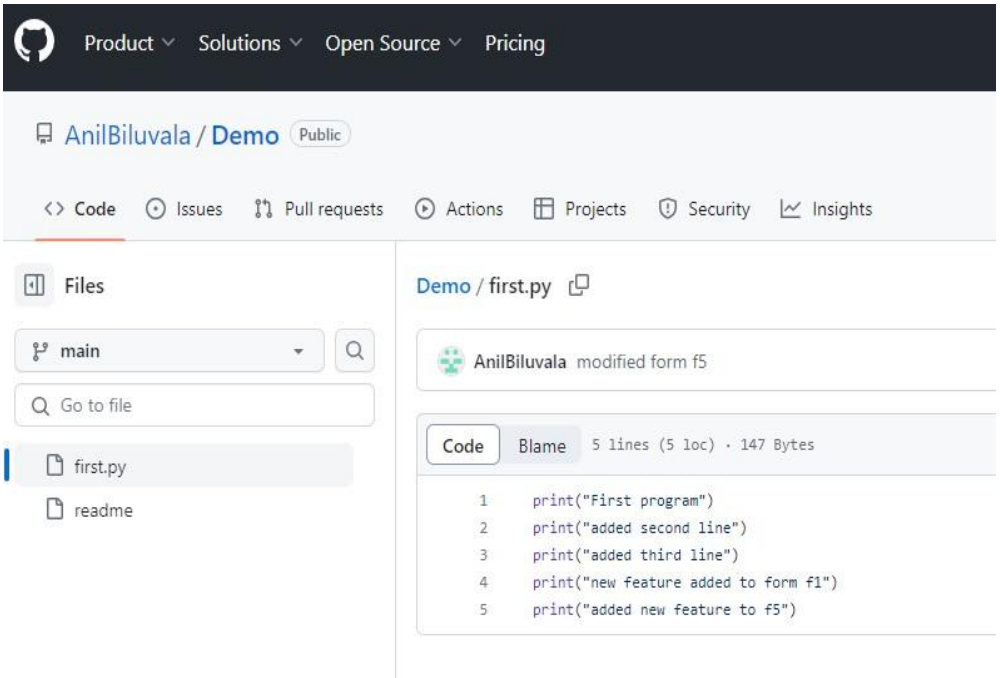
•Select build now. You can see a building history has been created. Click on that. In the console output, you can see the output of the first Jenkins job with time and date. Project configuration

1. Install python interpreter on local machine.

2. Write python script and debug the program for any errors.



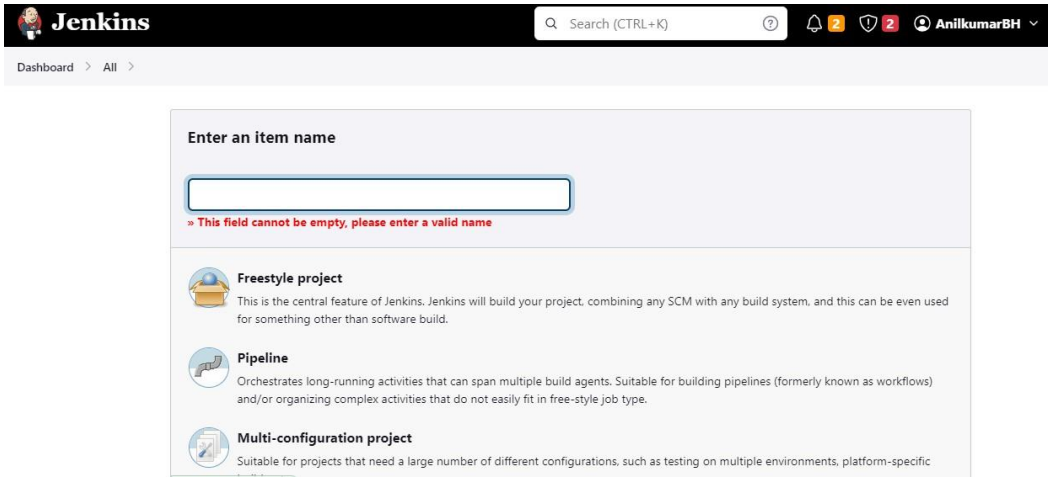
3. Push the project into git repository.



4. Launch the jenkins and configure the pipeline.

Configuration of Jenkins

1. Create new project and provide name for the project. Choose “Freestyle Project”.

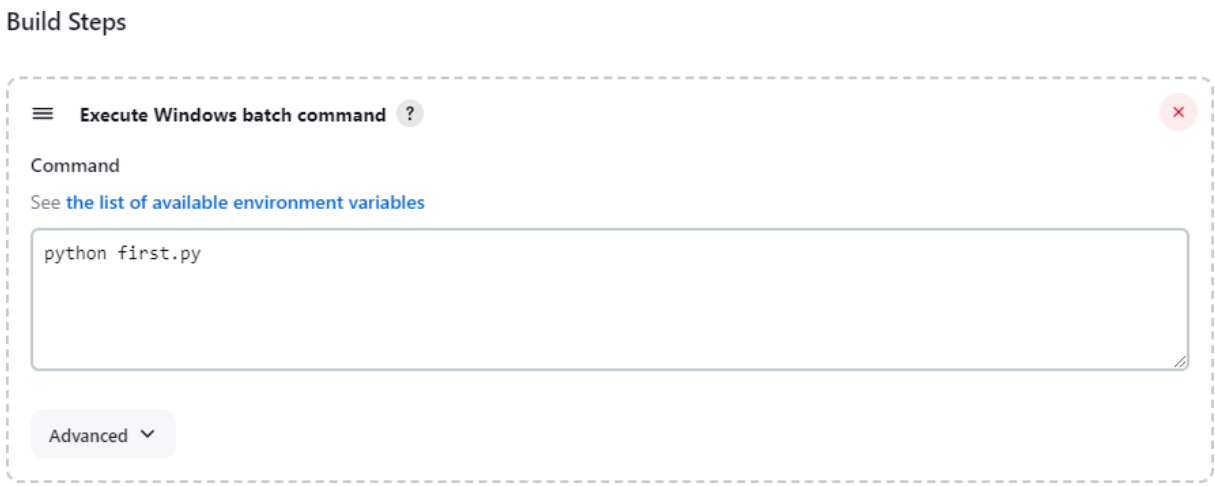


2. Click on newly created project and move to configuration. Under source Code Management provide git repository.

3. Provide branch name where project is located.

4. Under Build Triggers select Poll SCM and configure the schedule according to your project.

5. In build steps choose “Execute Windows batch command” and provide command for execution.



6. Apply and save your configuration.

7. Click on “Build Now” for detailed output click on “Console Output” and verify the process. Finally build status will be displayed according to your project implementation.

B-Exercise

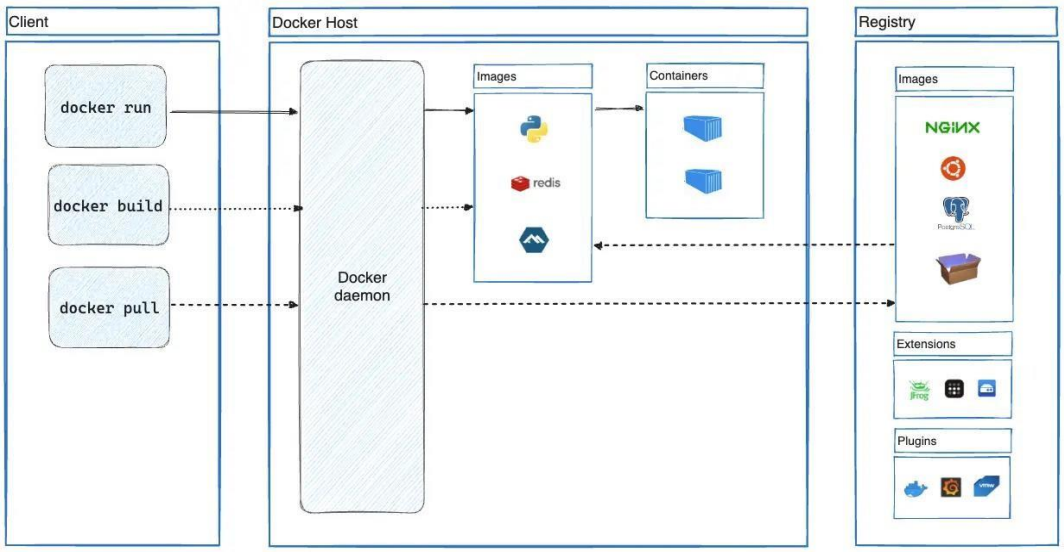
4. Create a docker image for an application stored in local repository and run the application using docker image.

Introduction to Docker

Docker is an open platform for developing, shipping, and running applications. Docker enables you to separate your applications from your infrastructure so you can deliver software quickly. With Docker, you can manage your infrastructure in the same ways you manage your applications. By taking advantage of Docker's methodologies for shipping, testing, and deploying code, you can significantly reduce the delay between writing code and running it in production.

Docker architecture

Docker uses a client-server architecture. The Docker client talks to the Docker daemon, which does the heavy lifting of building, running, and distributing your Docker containers. The Docker client and daemon can run on the same system, or you can connect a Docker client to a remote Docker daemon. The Docker client and daemon communicate using a REST API, over UNIX sockets or a network interface. Another Docker client is Docker Compose, that lets you work with applications consisting of a set of containers.



Install Docker Desktop on Windows

System requirements

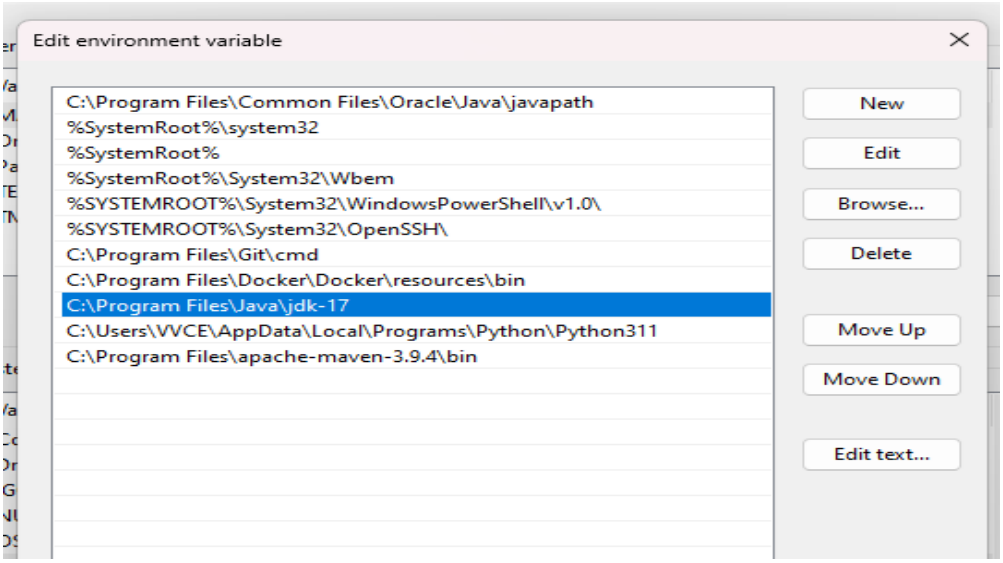
- WSL version 1.1.3.0 or later.
- Windows 11 64-bit: Home or Pro version 21H2 or higher, or Enterprise or Education version 21H2 or higher.
- Windows 10 64-bit:
- We recommend Home or Pro 22H2 (build 19045) or higher, or Enterprise or Education 22H2 (build 19045) or higher.
- Minimum required is Home or Pro 21H2 (build 19044) or higher, or Enterprise or Education 21H2 (build 19044) or higher.
- Turn on the WSL 2 feature on Windows. For detailed instructions, refer to the Microsoft documentationopen_in_new.

The following hardware prerequisites are required to successfully run WSL 2 on Windows 10 or Windows 11:

- 64-bit processor with Second Level Address Translation (SLAT)open_in_new
- 4GB system RAM
- Enable hardware virtualization in BIOS. For more information, see Virtualization. Configuration of Java applications

1. Install prerequisites for java applications like jdk tool.

2. Configure environment variables for jdk.



3. Create java application.

```
Calci.java   sample.java ×   Dockerfile
sample.java
1  public class sample {
2      public static void main(String args[]){
3          System.out.println("Test");
4
5      }
6
7  }
8
```

4. Create Dockerfile where configuration details are provided for creating image.

```
Calci.java   sample.java   Dockerfile ×
Dockerfile > ...
1  FROM openjdk
2  WORKDIR /app
3  COPY . /app
4  RUN javac Calci.java
5  CMD ["java","Calci"]
```

5. Complile the java program for any errors.

6. Create image for the application.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   powershell +

PS C:\Users\VVCE\Desktop\docker image> docker build -t program4 .
* History restored

PS C:\Users\VVCE\Desktop\docker image>
```

7. Run the application using docker image.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   powershell + ⌄

PS C:\Users\VVCE\Desktop\docker image> docker build -t program4 .
* History restored

PS C:\Users\VVCE\Desktop\docker image> docker run --name javaappli program4
```

5. **Create a process to dockerize the image for mysql database, demonstrate the working of client and server model using docker.**
   1. Run the docker in background and in command prompt type
      Docker run hello-world enter
   2. In browser search for hub,docker.com and login to store the images and containers then search for mysql and enter
   3. In another prompt for server setup type docker pull mysql and enter
   4. Docker run mysql
   5. Docker run -e MYSQL_ROOT_PASSWORD=admin -d mysql
   6. In 3rd prompt type docker run -it mysql /bin/bash enter and mysql -h enter
   7. Docker container ls
   8. Docker inspect paste any name of the any image
   9. Go to client terminal and create a table

6. **Create and configure Jenkins files for workflow and build of an application and push the image**
   - Download Jenkins 2.375.1 LTS for windows and jdk 17(Set the path in environment variable)
   - Select run service as local system. And click on next
   - Select test port and next
   - Paste the path of jdk file and select firewall exception will be installed after clicking next
   - In browser: localhost:8080 then enter
   - Give admin password where they have mentioned and create an account in jenkins

   Write a java program named “simple.java” and push to git hub and copy the link, then goto Jenkins click on new item.
   - Check the git hub project option.
   - In source code section paste simple.java program url.
   - In batch command give **java simple.java** save and then click on build.

7. **Create a maven project with all dependencies required for the application in CI/CD pipeline.**
    - create a maven project in eclipse.
    - Select filter as org.apache.maven.archetypes , maven. Archetype-quickstart 1.4 and then click on next.
    - Give group ID (name of the file) as ex1 article id as ex1 then click on finish.
    - In browser search for maven repo select simple json file 1.1.1 and copy the dependencies.
    - In eclipse ex1/src/main.java select pom.xml and paste the dependencies.
    - Create a new file called emp.json and type the simple program
    - { “fname”:”AI”, “lname”:”ML”} And save the program.
    - In src/test/java→ new→ class→ Name: readex and click on finish.
    - Run the program.
8. **Integrate communication channel with Jenkins for status of project and enable email notification for a build.**
    - Download slack for windows 64bit.
    - Create another workspace named as aiml →skip→aiml.
    - In Jenkins click on manage Jenkins→plugins→available plugins there select slack notification and global slack notifier and install.
    - After installing goto installed plugins→slack notification → select second url another window will open, there post the channel name(#aiml)
    - Then click on add Jenkins CI integration and copy the secrete key in step3.
    - Then goto manage Jenkins→system→slack→ give workspace name and add credential as Jenkins, in kind→ secret text there paste the secrete key give description and click on add . again click on credentials and select added description name then save.
    - Jenkins→ new item give batch command as java –version , build action→ slack notification (check all the options) and click on build. Notification will displayed in slack notification.

9. **Apply Xpath to locate elements in an application for automation testing in CI/CD pipeline**.

Java 8/above, jdk 18 and eclipse should be installed.

Selenium installation: latest stable version, download browser driver and configure selenium on your eclipse

Selenium.org→ downloads select 4.3.0

Create a folder called selenium webdriver in C drive then extract and paste the downloaded selenium driver jar file.

To Install chrome driver : select chrome driver (same version as your chrome browser) and download.

Create a folder called webdriver in C drive then extract and paste the downloaded file

In eclipse create a new java project give name seledemo click on use an execution environment JRE: javaSe18→ finish

To create package seledemo right click→new→package give name as ex

Src right click→new class(sescript)→ select public static void main and write a simple program

To include selenium file:

Src→ build path→ configure build path → libraries→add external JASR file files paste the path link of selenium webdriver folder in C drive.

```
public static void main(String[] args)
{
System.setProperty("path of chromedriver in C drive ");
WebDriver driver = new chromeDriver();
driver.get("vvce.ac.in");
System.out.println(driver.getTitle());
Driver.quit();
}
```

VIVA Questions

1. What is devops?
2. What is a DevOps Engineer?
3. Which are some of the most popular DevOps tools?
4. How is DevOps different from agile methodology?
5. What are the different phases in DevOps?
6. Mention some of the core benefits of DevOps
7. What is the difference between continuous delivery and continuous deployment?
8. What is Jenkins?
9. Explain how you can clone a Git repository via Jenkins?
10. Explain how you can set up Jenkins job?
11. Mention what are the two components Jenkins is mainly integrated with?
12. What is Git and why is it used?
13. What is a version control system (VCS)?
14. What is a git repository?
15. What does git clone do?
16. What does the command git config do?
17. What is a conflict?
18. What does git status command do?
19. What does git add command do?
20. How will you create a git repository?
21. What do the git diff and git status commands do?
22. What is the command used to delete a branch?
23. Difference between git pull and git fetch
24. difference between git and git hub
25. What do the git diff and git status commands do?